system/ Purpose + boundaries

Purpose

Piece A — Purpose (what system/ is)

File: `Axion/libraries/system/SYS-0_purpose.md`

---
library: system
id: SYS-0
schema_version: 1.0.0
status: draft

SYS-0 — system/ Purpose + Boundaries

Purpose
`system/` is the **control-plane configuration and runtime contracts** for Axion. It defines the stable “operating environment” that every run depends on:
- how workspaces/projects are represented
- how pins/locks are applied for deterministic resolution
- how adapters declare capabilities (local/Replit/CI)
- how quotas and rate limits are enforced
- how notifications are routed
- how policy engine hooks are invoked
- how system-level state and operator actions are represented

What it governs (in scope)
- **Workspace / project model**: namespaces, org/project boundaries, permissions, keys
- **Pin + lock policies**: rules for selecting and freezing library versions, standards snapshots, templates, knowledge selections
- **Adapter manager**: runtime environment capability discovery (what tools/commands are allowed where)
- **Quota + rate limits**: per project/profile constraints on runs, tokens, storage, compute
- **Notification routing**: gate failures, approvals, policy violations, run status events
- **System-level state contracts**: stable identifiers and required metadata for system actions

What it does NOT govern (out of scope)
- Pipeline stage order and stage IO contracts → `orchestration/`
- Gate DSL language and evaluation semantics → `gates/`
- Risk classes and override policies (the meaning of “prototype/prod/compliance”) → `policy/`
- Intake forms and submission normalization → `intake/`
- Canonical spec structure and unknown handling → `canonical/`
- Standards packs content and standards resolution → `standards/`
- Templates library + rendering rules → `templates/`
- Knowledge library content, taxonomy, selection rules → `knowledge/`

- Proof ledger and verification completion criteria → `verification/`
- Kit packaging structure and export rules → `kit/`
- Telemetry event schemas and sinks → `telemetry/`
- Ops standards (SLO/alerts/logging budgets) → `ops/`
- Audit ledger rules (operator action tracking) → `audit/`

Consumers (what reads system/)
- Control plane runtime (orchestrator + state machine)
- Gate evaluator (for policy hooks and notification emission)
- Adapter manager (capability discovery and enforcement)
- Run launcher (quota checks, pins, workspace permissions)

Determinism and stability requirements
- IDs are stable and not reused.
- Pins/locks are explicit and recorded in the run manifest.
- Capability and quota policies are resolved deterministically from:
  workspace → project → profile → run context.

Outputs (what system/ produces)
- System configuration snapshots used at runtime
- Enforcement decisions (allowed/blocked actions) referenced in run logs and gate reports
- Notification events for operators

Boundary checklist

Piece B — Boundary checklist (to prevent drift later)

File: `Axion/libraries/system/SYS-0_boundary_checklist.md`

---
library: system
id: SYS-0a
schema_version: 1.0.0
status: draft

SYS-0a — Boundary Checklist

A change belongs in `system/` if it answers at least one:
- “What environment is this run allowed to execute in?”
- “What resources is this run allowed to consume?”
- “What versions are pinned/locked and how?”
- “What notifications should fire and where?”
- “What capabilities does this adapter/runtime have?”

A change does NOT belong in `system/` if it answers:
- “What does the pipeline do in stage Sx?” (orchestration)
- “How do gates evaluate predicates?” (gates)
- “What risk class rules exist?” (policy)
- “What template/knowledge/standards content exists?” (those libraries)

Workspace / project model

Piece A — Concept + required entities

File: `Axion/libraries/system/SYS-1_workspace_project_model.md`

---
library: system
id: SYS-1
schema_version: 1.0.0
status: draft

SYS-1 — Workspace / Project Model

Purpose
Define the canonical structure for organizing Axion usage into stable namespaces with predictable permissions, quotas, and pin policies.

Core entities
- **Workspace**: top-level namespace (usually an org or environment boundary)
- **Project**: runnable unit inside a workspace (runs, templates, policies, and pins attach here)
- **Profile**: execution profile for a run context (API/Web/Mobile/etc.), used for quotas and selection

Required properties (high level)
Workspace must have:
- workspace_id (stable)
- name
- owner
- created_at, updated_at
- default_policies (pins/locks baseline, quota baseline)

Project must have:
- project_id (stable)
- workspace_id (parent)
- name
- owners / maintainers
- created_at, updated_at
- run_profiles enabled
- policy bindings (risk defaults, overrides permissions)
- quota bindings
- pin/lock bindings (what can be pinned, who can pin)

Profile must have:

- profile_id (stable, enumerated)
- name
- capability requirements (adapter constraints)
- quota modifiers (optional)

Piece B — Machine-readable schema (workspace)

File: `Axion/libraries/system/schemas/workspace.v1.schema.json`

```
{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "schema_version": "1.0.0",
 "id": "workspace.v1",
 "type": "object",
 "required": [
   "workspace_id",
   "name",
   "owner",
   "created_at",
   "updated_at",
   "defaults"
 ],
 "properties": {
   "workspace_id": { "type": "string", "pattern": "^WS-[A-Z0-9]{6,}$" },
   "name": { "type": "string", "minLength": 2 },
   "owner": { "type": "string", "minLength": 2 },
   "created_at": { "type": "string", "format": "date-time" },
   "updated_at": { "type": "string", "format": "date-time" },

   "defaults": {
     "type": "object",
     "required": ["policy_set_id", "quota_set_id", "pin_policy_id"],
     "properties": {
       "policy_set_id": { "type": "string", "minLength": 3 },
       "quota_set_id": { "type": "string", "minLength": 3 },
       "pin_policy_id": { "type": "string", "minLength": 3 }
     },
     "additionalProperties": false
   },

   "labels": {
     "type": "object",
     "additionalProperties": { "type": "string" }
```

```
  }
 },
 "additionalProperties": false
}
```

Piece C — Machine-readable schema (project)

File: `Axion/libraries/system/schemas/project.v1.schema.json`

```
{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "schema_version": "1.0.0",
 "id": "project.v1",
 "type": "object",
 "required": [
   "project_id",
   "workspace_id",
   "name",
   "owners",
   "created_at",
   "updated_at",
   "enabled_profiles",
   "bindings"
 ],
 "properties": {
   "project_id": { "type": "string", "pattern": "^PRJ-[A-Z0-9]{6,}$" },
   "workspace_id": { "type": "string", "pattern": "^WS-[A-Z0-9]{6,}$" },
   "name": { "type": "string", "minLength": 2 },

   "owners": { "type": "array", "minItems": 1, "items": { "type": "string" } },
   "maintainers": { "type": "array", "items": { "type": "string" } },

   "created_at": { "type": "string", "format": "date-time" },
   "updated_at": { "type": "string", "format": "date-time" },

   "enabled_profiles": {
     "type": "array",
     "minItems": 1,
     "items": { "type": "string", "pattern": "^PROFILE-[A-Z0-9_]+$" }
   },

   "bindings": {
     "type": "object",
```

```
    "required": ["policy_set_id", "quota_set_id", "pin_policy_id"],
    "properties": {
      "policy_set_id": { "type": "string", "minLength": 3 },
      "quota_set_id": { "type": "string", "minLength": 3 },
      "pin_policy_id": { "type": "string", "minLength": 3 }
    },
    "additionalProperties": false
  },

  "labels": {
    "type": "object",
    "additionalProperties": { "type": "string" }
  }
},
"additionalProperties": false
}
```

Piece D — Profile registry (enumerated profiles)

File: `Axion/libraries/system/registries/run_profiles.v1.json`

```
{
 "schema_version": "1.0.0",
 "registry_id": "run_profiles.v1",
 "profiles": [
   { "profile_id": "PROFILE-API", "name": "API" },
   { "profile_id": "PROFILE-WEB", "name": "Web App" },
   { "profile_id": "PROFILE-MOBILE", "name": "Mobile App" },
   { "profile_id": "PROFILE-INFRA", "name": "Infrastructure" },
   { "profile_id": "PROFILE-DATA", "name": "Data" },
   { "profile_id": "PROFILE-MIXED", "name": "Mixed" }
 ]
}
```

Piece E — Determinism rules

File: `Axion/libraries/system/SYS-1_determinism_rules.md`

```
---
library: system
id: SYS-1a
schema_version: 1.0.0
```

status: draft

SYS-1a — Determinism Rules (Workspace/Project)

- workspace_id and project_id are immutable.
- A project must reference exactly one parent workspace_id.
- Effective policy/quota/pin settings are resolved in this order:
 1) Workspace defaults
 2) Project bindings override workspace defaults
 3) Run-level pins (if allowed) override project bindings (recorded in run manifest)
- enabled_profiles gate what runs can be launched for the project.

Piece F — Validation checklist

File: `Axion/libraries/system/SYS-1_validation_checklist.md`

---
library: system
id: SYS-1b
schema_version: 1.0.0
status: draft

SYS-1b — Validation Checklist

- [ ] workspace record matches workspace.v1 schema
- [ ] project record matches project.v1 schema
- [ ] project.workspace_id points to an existing workspace_id
- [ ] enabled_profiles entries exist in run_profiles.v1
- [ ] bindings include policy_set_id, quota_set_id, pin_policy_id

Pin/lock policies

SYS-2 — Pin / lock policies (deterministic resolution)

Piece A — Concept and scope

File: `Axion/libraries/system/SYS-2_pin_lock_policies.md`

---
library: system
id: SYS-2
schema_version: 1.0.0
status: draft

SYS-2 — Pin / Lock Policies

Purpose
Pins/locks make runs reproducible by freezing which library assets are used (schemas, policies, standards snapshots, templates, knowledge bundles, etc.).

Definitions
- **Pin**: an explicit reference to a specific version of a library asset.
- **Lock**: an enforcement rule that forbids changing a pinned asset (or forbids “latest” resolution).
- **Resolution**: mapping from logical asset name → concrete path + version.

What can be pinned (pin targets)
Minimum pin targets:
- library_index version (global library resolution)
- schema_registry version (schema resolution)
- standards snapshot id
- template registry version + selected template ids
- knowledge selection artifact id (or bundle ids + resolved KID list)
- policy set id
- quota set id
- adapter profile id (capabilities baseline)

Where pins live
Pins exist at three levels:
1) Workspace defaults (baseline pins)
2) Project bindings (override pins)
3) Run manifest pins (final, per-run)

Determinism rule
A run is reproducible if its run manifest contains the fully resolved pins for all required targets.

Piece B — Pin policy schema (what pins are allowed + who can set them)

File: `Axion/libraries/system/schemas/pin_policy.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "pin_policy.v1",
  "type": "object",
  "required": ["pin_policy_id", "rules", "created_at", "updated_at", "owner"],
  "properties": {
    "pin_policy_id": { "type": "string", "pattern": "^PINPOL-[A-Z0-9]{6,}$" },

    "rules": {
      "type": "object",
      "required": ["allowed_targets", "lock_mode", "who_can_pin"],
      "properties": {
        "allowed_targets": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": "string",
            "enum": [
              "library_index",
              "schema_registry",
              "standards_snapshot",
              "template_registry",
              "template_selection",
              "knowledge_selection",
              "policy_set",
              "quota_set",
              "adapter_profile"
            ]
          }
        },

        "lock_mode": {
          "type": "string",
          "enum": ["open", "pinned_required", "locked"],
```

```
        "description": "open: allow latest; pinned_required: must pin at run start; locked: only pinned versions allowed, no changes"
      },

      "who_can_pin": {
        "type": "array",
        "items": { "type": "string", "enum": ["owner", "maintainer", "operator"] },
        "minItems": 1
      },

      "allow_run_level_overrides": { "type": "boolean", "default": false }
    },
    "additionalProperties": false
  },

  "created_at": { "type": "string", "format": "date-time" },
  "updated_at": { "type": "string", "format": "date-time" },
  "owner": { "type": "string", "minLength": 2 }
},
"additionalProperties": false
}
```

Piece C — Pin set schema (the resolved pins recorded in a run)

File: `Axion/libraries/system/schemas/pin_set.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "pin_set.v1",
  "type": "object",
  "required": ["pin_set_id", "pins", "resolved_at"],
  "properties": {
    "pin_set_id": { "type": "string", "pattern": "^PINSET-[A-Z0-9]{6,}$" },
    "resolved_at": { "type": "string", "format": "date-time" },

    "pins": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "object",
        "required": ["target", "ref"],
```

```
      "properties": {
        "target": {
          "type": "string",
          "enum": [
            "library_index",
            "schema_registry",
            "standards_snapshot",
            "template_registry",
            "template_selection",
            "knowledge_selection",
            "policy_set",
            "quota_set",
            "adapter_profile"
          ]
        },
        "ref": {
          "type": "object",
          "required": ["version", "path"],
          "properties": {
            "version": { "type": "string", "minLength": 1 },
            "path": { "type": "string", "minLength": 3 },
            "hash": { "type": "string" }
          },
          "additionalProperties": false
        }
      },
      "additionalProperties": false
    }
  }
},
"additionalProperties": false
}
```

Piece D — Resolution rules (precedence + fill-in requirements)

File: `Axion/libraries/system/SYS-2_resolution_rules.md`

---
library: system
id: SYS-2a
schema_version: 1.0.0
status: draft

SYS-2a — Pin Resolution Rules

Precedence
When resolving pins for a run:
1) Start from Workspace defaults
2) Apply Project bindings (override)
3) Apply Run-level overrides (only if pin_policy allows)

Lock modes
- open:
  - allowed to resolve “latest”
  - still records resolved versions into pin_set at run start
- pinned_required:
  - run cannot start unless all required targets are pinned
- locked:
  - run cannot start unless pins match allowed pinned versions
  - “latest” is forbidden

Completion requirement
Before stage execution begins, the run must produce a resolved pin_set where:
- every required target has a concrete (version, path, optional hash)
- the pin_set is recorded in the run manifest

Piece E — Validation checklist

File: `Axion/libraries/system/SYS-2_validation_checklist.md`

---
library: system
id: SYS-2b
schema_version: 1.0.0
status: draft

SYS-2b — Validation Checklist

- [ ] pin_policy matches pin_policy.v1 schema
- [ ] pin_set matches pin_set.v1 schema
- [ ] pin_set includes all required targets for the run profile
- [ ] lock_mode rules enforced:

  - [ ] pinned_required blocks missing pins
  - [ ] locked blocks latest/unpinned resolution
- [ ] resolved pin_set recorded in run manifest before execution

Adapter manager

SYS-3 — Adapter manager (envs + capability discovery)

Piece A — Concept and scope

File: `Axion/libraries/system/SYS-3_adapter_manager.md`

---
library: system
id: SYS-3
schema_version: 1.0.0
status: draft

SYS-3 — Adapter Manager (Capability Discovery)

Purpose
Adapters define **where** a run executes (local, Replit, CI, etc.) and **what it is allowed to do** there. The adapter manager provides deterministic capability discovery so agents can't assume tools that don't exist.

Definitions
- **Adapter**: an execution environment integration (Local, Replit, GitHub Actions, etc.)
- **Capability**: a declared, auditable ability (run commands, write files, access network, run tests, etc.)
- **Adapter profile**: a named bundle of capabilities used for pinning and enforcement

What it governs
- Capability declarations (what exists)
- Capability requirements per run profile (API/Web/Mobile/etc.)
- Command policy boundaries (what commands are allowed)
- Environmental constraints (no network, limited ports, sandbox rules)

What it does NOT govern
- The content of verification commands (that's `verification/`)
- Stage order/IO (that's `orchestration/`)
- Risk class policies (that's `policy/`)

Piece B — Capability registry schema

File: `Axion/libraries/system/schemas/capability_registry.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "capability_registry.v1",
  "type": "object",
  "required": ["registry_id", "schema_version", "capabilities"],
  "properties": {
    "registry_id": { "type": "string", "minLength": 3 },
    "schema_version": { "type": "string", "minLength": 1 },
    "capabilities": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "object",
        "required": ["capability_id", "name", "category"],
        "properties": {
          "capability_id": { "type": "string", "pattern": "^CAP-[A-Z0-9_]+$" },
          "name": { "type": "string", "minLength": 2 },
          "category": {
            "type": "string",
            "enum": ["filesystem", "process", "network", "runtime", "secrets", "ui", "git"]
          },
          "notes": { "type": "string" }
        },
        "additionalProperties": false
      }
    }
  },
  "additionalProperties": false
}
```

Piece C — Adapter profile schema (bundle of capabilities)

File: `Axion/libraries/system/schemas/adapter_profile.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "adapter_profile.v1",
  "type": "object",
  "required": ["adapter_profile_id", "adapter_type", "capabilities", "command_policy_id",
"created_at", "updated_at", "owner"],
```

```
  "properties": {
    "adapter_profile_id": { "type": "string", "pattern": "^ADP-[A-Z0-9]{6,}$" },
    "adapter_type": {
      "type": "string",
      "enum": ["local", "replit", "ci", "container"]
    },
    "capabilities": {
      "type": "array",
      "minItems": 1,
      "items": { "type": "string", "pattern": "^CAP-[A-Z0-9_]+$" }
    },
    "command_policy_id": { "type": "string", "minLength": 3 },
    "constraints": {
      "type": "object",
      "properties": {
        "network_access": { "type": "string", "enum": ["none", "restricted", "full"] },
        "allowed_ports": { "type": "array", "items": { "type": "integer", "minimum": 1, "maximum":
65535 } },
        "max_processes": { "type": "integer", "minimum": 1 },
        "max_runtime_minutes": { "type": "integer", "minimum": 1 }
      },
      "additionalProperties": false
    },
    "created_at": { "type": "string", "format": "date-time" },
    "updated_at": { "type": "string", "format": "date-time" },
    "owner": { "type": "string", "minLength": 2 }
  },
  "additionalProperties": false
}
```

Piece D — Command policy schema (allowed command patterns)

File: `Axion/libraries/system/schemas/command_policy.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "command_policy.v1",
  "type": "object",
  "required": ["command_policy_id", "allow", "deny", "created_at", "updated_at", "owner"],
  "properties": {
    "command_policy_id": { "type": "string", "pattern": "^CMDPOL-[A-Z0-9]{6,}$" },
```

```
    "allow": {
      "type": "array",
      "items": { "type": "string", "minLength": 1 },
      "description": "Allowed command patterns (simple glob/regex patterns; engine-defined)"
    },
    "deny": {
      "type": "array",
      "items": { "type": "string", "minLength": 1 },
      "description": "Denied command patterns take precedence"
    },
    "created_at": { "type": "string", "format": "date-time" },
    "updated_at": { "type": "string", "format": "date-time" },
    "owner": { "type": "string", "minLength": 2 }
  },
  "additionalProperties": false
}
```

Piece E — Default capability set (starter registry)

File: `Axion/libraries/system/registries/capabilities.v1.json`

```
{
  "schema_version": "1.0.0",
  "registry_id": "capabilities.v1",
  "capabilities": [
    { "capability_id": "CAP-FS_READ", "name": "Read files", "category": "filesystem" },
    { "capability_id": "CAP-FS_WRITE", "name": "Write files", "category": "filesystem" },
    { "capability_id": "CAP-PROC_SPAWN", "name": "Spawn processes", "category": "process" },
    { "capability_id": "CAP-NET_NONE", "name": "No network access", "category": "network" },
    { "capability_id": "CAP-NET_RESTRICTED", "name": "Restricted network access",
"category": "network" },
    { "capability_id": "CAP-GIT_READ", "name": "Read git repo", "category": "git" },
    { "capability_id": "CAP-GIT_WRITE", "name": "Write git repo", "category": "git" },
    { "capability_id": "CAP-SECRETS_READ", "name": "Read injected secrets", "category":
"secrets" }
  ]
}
```

Piece F — Determinism rules

File: `Axion/libraries/system/SYS-3_determinism_rules.md`

```
---
library: system
id: SYS-3a
schema_version: 1.0.0
status: draft
---

# SYS-3a — Determinism Rules (Adapters)

- Runs must pin an adapter_profile_id (SYS-2 pin target: adapter_profile).
- Adapter profiles must reference capabilities by ID from capabilities registry.
- If capabilities differ between environments, the adapter profile must reflect that explicitly.
- Command policy deny rules override allow rules.
- Capability resolution must be recorded in the run manifest (adapter_profile snapshot).
```

Piece G — Validation checklist

File: `Axion/libraries/system/SYS-3_validation_checklist.md`

```
---
library: system
id: SYS-3b
schema_version: 1.0.0
status: draft
---

# SYS-3b — Validation Checklist

- [ ] capabilities registry matches capability_registry schema
- [ ] adapter profile matches adapter_profile schema
- [ ] adapter profile capability IDs exist in capabilities registry
- [ ] command policy matches command_policy schema
- [ ] deny patterns override allow patterns
- [ ] run manifest pins adapter_profile_id and records resolved snapshot
```

.

Quotas + rate limits

SYS-4 — Quotas + rate limits (per project/profile)

Piece A — Concept and scope

File: `Axion/libraries/system/SYS-4_quotas_rate_limits.md`

```
---
library: system
id: SYS-4
schema_version: 1.0.0
status: draft
---

# SYS-4 — Quotas + Rate Limits

## Purpose
Define deterministic limits on resource usage so Axion runs stay controllable:
- runs/day, concurrent runs
- token budgets (LLM usage)
- compute/runtime ceilings
- storage ceilings
- outbound network limits (if allowed)

Quotas are resolved from workspace → project → profile → run context.

## Definitions
- **Quota set**: a named set of limits.
- **Rate limit**: a time-based throttle (per minute/hour/day).
- **Budget**: a cap on a measured unit (tokens, minutes, MB).
- **Enforcement action**: block, warn, degrade, or require approval.

## What it governs
- quota sets used by workspaces/projects (bindings)
- profile modifiers (API vs MOBILE can have different budgets)
- deterministic enforcement decisions recorded in run manifest
```

Piece B — Quota set schema

File: `Axion/libraries/system/schemas/quota_set.v1.schema.json`

```
{
```

```
"$schema": "https://json-schema.org/draft/2020-12/schema",
"schema_version": "1.0.0",
"id": "quota_set.v1",
"type": "object",
"required": ["quota_set_id", "limits", "enforcement", "created_at", "updated_at", "owner"],
"properties": {
  "quota_set_id": { "type": "string", "pattern": "^QUOTA-[A-Z0-9]{6,}$" },

  "limits": {
    "type": "object",
    "required": ["runs", "tokens", "compute", "storage"],
    "properties": {
      "runs": {
        "type": "object",
        "required": ["per_day", "concurrent_max"],
        "properties": {
          "per_day": { "type": "integer", "minimum": 0 },
          "concurrent_max": { "type": "integer", "minimum": 1 }
        },
        "additionalProperties": false
      },

      "tokens": {
        "type": "object",
        "required": ["per_run_max", "per_day_max"],
        "properties": {
          "per_run_max": { "type": "integer", "minimum": 0 },
          "per_day_max": { "type": "integer", "minimum": 0 }
        },
        "additionalProperties": false
      },

      "compute": {
        "type": "object",
        "required": ["runtime_minutes_max"],
        "properties": {
          "runtime_minutes_max": { "type": "integer", "minimum": 1 }
        },
        "additionalProperties": false
      },

      "storage": {
        "type": "object",
        "required": ["artifacts_mb_max", "kits_mb_max"],
```

```
        "properties": {
          "artifacts_mb_max": { "type": "integer", "minimum": 0 },
          "kits_mb_max": { "type": "integer", "minimum": 0 }
        },
        "additionalProperties": false
      },

      "network": {
        "type": "object",
        "properties": {
          "egress_mb_max": { "type": "integer", "minimum": 0 },
          "requests_per_minute_max": { "type": "integer", "minimum": 0 }
        },
        "additionalProperties": false
      }
    },
    "additionalProperties": false
  },

  "enforcement": {
    "type": "object",
    "required": ["on_exceed"],
    "properties": {
      "on_exceed": {
        "type": "string",
        "enum": ["block", "warn", "require_approval", "degrade"]
      },
      "degrade_mode": {
        "type": "string",
        "enum": ["reduce_selection_cap", "reduce_template_scope", "disable_optional_stages"],
        "description": "Only used when on_exceed == degrade"
      }
    },
    "additionalProperties": false
  },

  "created_at": { "type": "string", "format": "date-time" },
  "updated_at": { "type": "string", "format": "date-time" },
  "owner": { "type": "string", "minLength": 2 }
 },
 "additionalProperties": false
}
```

Piece C — Profile quota modifiers schema (optional)

File: `Axion/libraries/system/schemas/quota_profile_modifiers.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "quota_profile_modifiers.v1",
  "type": "object",
  "required": ["registry_id", "modifiers"],
  "properties": {
    "registry_id": { "type": "string", "minLength": 3 },
    "modifiers": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["profile_id", "multiplier"],
        "properties": {
          "profile_id": { "type": "string", "pattern": "^PROFILE-[A-Z0-9_]+$" },
          "multiplier": {
            "type": "object",
            "properties": {
              "tokens_per_run": { "type": "number", "minimum": 0 },
              "runtime_minutes": { "type": "number", "minimum": 0 }
            },
            "additionalProperties": false
          }
        },
        "additionalProperties": false
      }
    }
  },
  "additionalProperties": false
}
```

Piece D — Starter quota set (example baseline)

File: `Axion/libraries/system/registries/quota_sets.v1.json`

```
{
  "schema_version": "1.0.0",
```

```
  "registry_id": "quota_sets.v1",
  "quota_sets": [
    {
      "quota_set_id": "QUOTA-BASE01",
      "limits": {
        "runs": { "per_day": 20, "concurrent_max": 2 },
        "tokens": { "per_run_max": 250000, "per_day_max": 1500000 },
        "compute": { "runtime_minutes_max": 60 },
        "storage": { "artifacts_mb_max": 500, "kits_mb_max": 500 }
      },
      "enforcement": { "on_exceed": "block" },
      "created_at": "2026-03-05T00:00:00Z",
      "updated_at": "2026-03-05T00:00:00Z",
      "owner": "axion"
    }
  ]
}
```

Piece E — Resolution rules (how effective limits are computed)

File: `Axion/libraries/system/SYS-4_resolution_rules.md`

```
---
library: system
id: SYS-4a
schema_version: 1.0.0
status: draft
---

# SYS-4a — Quota Resolution Rules

## Inputs
- workspace.defaults.quota_set_id
- project.bindings.quota_set_id
- run profile_id
- optional quota_profile_modifiers

## Precedence
1) Workspace default quota_set_id
2) Project binding overrides workspace
3) Apply profile multipliers (if present)
4) Record final effective quota in run manifest
```

Enforcement decision
- If a limit would be exceeded at run start:
  apply enforcement.on_exceed (block/warn/require_approval/degrade)
- Enforcement decisions must be recorded:
  - decision
  - reason (which limit)
  - computed effective limits

Piece F — Validation checklist

File: `Axion/libraries/system/SYS-4_validation_checklist.md`

---
library: system
id: SYS-4b
schema_version: 1.0.0
status: draft

SYS-4b — Validation Checklist

- [ ] quota_set matches quota_set.v1 schema
- [ ] project/workspace quota_set_id references an existing quota set (strict mode)
- [ ] profile modifiers reference valid profile_id (run_profiles.v1)
- [ ] effective quotas are recorded in run manifest
- [ ] enforcement action applied deterministically when limits exceeded

Notification routing

SYS-5 — Notification routing (gate fail, approvals, standards/template changes)

Piece A — Concept and scope

File: `Axion/libraries/system/SYS-5_notification_routing.md`

```
---
library: system
id: SYS-5
schema_version: 1.0.0
status: draft
---

# SYS-5 — Notification Routing

## Purpose
Define how Axion emits notifications deterministically for:
- gate failures
- approvals required (policy overrides, quota exceed, risky actions)
- changes in pinned dependencies (standards/templates/knowledge)
- run lifecycle events (start, complete, failed, released)

## Definitions
- **Event**: a structured notification trigger produced by the pipeline/control plane.
- **Route**: mapping from event type + severity + context → destination(s).
- **Destination**: where the notification goes (console, email, Slack, webhook, etc.).
- **Throttle**: rules to prevent spam (dedupe/windowing).

## What it governs
- Event type registry
- Routing rules registry
- Destination registry
- Dedupe/throttle policy
- Deterministic rendering of a notification payload (title/body fields)

## What it does NOT govern
- How gates are evaluated (gates/)
- What policies mean (policy/)
- Telemetry sink policies (telemetry/)
```

Piece B — Event type registry schema

File:
`Axion/libraries/system/schemas/notification_event_types.v1.schema.json`

```
{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "schema_version": "1.0.0",
 "id": "notification_event_types.v1",
 "type": "object",
 "required": ["registry_id", "schema_version", "event_types"],
 "properties": {
   "registry_id": { "type": "string", "minLength": 3 },
   "schema_version": { "type": "string", "minLength": 1 },
   "event_types": {
     "type": "array",
     "minItems": 1,
     "items": {
       "type": "object",
       "required": ["event_type", "severity_levels"],
       "properties": {
         "event_type": { "type": "string", "pattern": "^[A-Z0-9_]+$" },
         "severity_levels": {
           "type": "array",
           "minItems": 1,
           "items": { "type": "string", "enum": ["info", "warning", "error", "critical"] }
         },
         "notes": { "type": "string" }
       },
       "additionalProperties": false
     }
   }
 },
 "additionalProperties": false
}
```

Piece C — Destination registry schema

File:
`Axion/libraries/system/schemas/notification_destinations.v1.schema.jso`
`n`

```
{
```

```
"$schema": "https://json-schema.org/draft/2020-12/schema",
"schema_version": "1.0.0",
"id": "notification_destinations.v1",
"type": "object",
"required": ["registry_id", "schema_version", "destinations"],
"properties": {
  "registry_id": { "type": "string", "minLength": 3 },
  "schema_version": { "type": "string", "minLength": 1 },
  "destinations": {
    "type": "array",
    "minItems": 1,
    "items": {
      "type": "object",
      "required": ["destination_id", "kind", "enabled"],
      "properties": {
        "destination_id": { "type": "string", "pattern": "^DEST-[A-Z0-9]{4,}$" },
        "kind": {
          "type": "string",
          "enum": ["console", "email", "slack", "webhook"]
        },
        "enabled": { "type": "boolean" },

        "config": {
          "type": "object",
          "description": "Kind-specific config; keep minimal here (secrets live elsewhere).",
          "properties": {
            "channel": { "type": "string" },
            "to": { "type": "array", "items": { "type": "string" } },
            "url_ref": { "type": "string", "description": "Reference to a secret, not a raw URL." }
          },
          "additionalProperties": false
        }
      },
      "additionalProperties": false
    }
  }
},
"additionalProperties": false
}
```

Piece D — Routing rules schema (event → destinations + throttle)

File: `Axion/libraries/system/schemas/notification_routes.v1.schema.json`

```
{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "schema_version": "1.0.0",
 "id": "notification_routes.v1",
 "type": "object",
 "required": ["routes_id", "schema_version", "routes"],
 "properties": {
   "routes_id": { "type": "string", "minLength": 3 },
   "schema_version": { "type": "string", "minLength": 1 },

   "routes": {
     "type": "array",
     "minItems": 1,
     "items": {
       "type": "object",
       "required": ["route_id", "match", "deliver", "throttle"],
       "properties": {
         "route_id": { "type": "string", "pattern": "^ROUTE-[A-Z0-9]{4,}$" },

         "match": {
           "type": "object",
           "required": ["event_type"],
           "properties": {
             "event_type": { "type": "string" },
             "severity": { "type": "array", "items": { "type": "string" } },
             "risk_class": { "type": "array", "items": { "type": "string" } },
             "profile_id": { "type": "array", "items": { "type": "string" } }
           },
           "additionalProperties": false
         },

         "deliver": {
           "type": "object",
           "required": ["destination_ids"],
           "properties": {
             "destination_ids": {
               "type": "array",
               "minItems": 1,
               "items": { "type": "string", "pattern": "^DEST-[A-Z0-9]{4,}$" }
             }
           },
           "additionalProperties": false
         },
```

```
        "throttle": {
          "type": "object",
          "required": ["mode"],
          "properties": {
            "mode": { "type": "string", "enum": ["none", "dedupe_window"] },
            "window_seconds": { "type": "integer", "minimum": 1 },
            "dedupe_key": {
              "type": "string",
              "enum": ["event_type", "event_type_plus_run", "event_type_plus_gate"]
            }
          },
          "additionalProperties": false
        }
      },
      "additionalProperties": false
    }
  }
 },
 "additionalProperties": false
}
```

Piece E — Starter registries (minimal runtime set)

File: `Axion/libraries/system/registries/notification_event_types.v1.json`

```
{
 "schema_version": "1.0.0",
 "registry_id": "notification_event_types.v1",
 "event_types": [
   { "event_type": "RUN_STARTED", "severity_levels": ["info"] },
   { "event_type": "RUN_COMPLETED", "severity_levels": ["info"] },
   { "event_type": "RUN_FAILED", "severity_levels": ["error", "critical"] },

   { "event_type": "GATE_FAILED", "severity_levels": ["warning", "error", "critical"] },
   { "event_type": "APPROVAL_REQUIRED", "severity_levels": ["warning", "error"] },

   { "event_type": "PIN_CHANGED", "severity_levels": ["warning"] },
   { "event_type": "STANDARDS_CHANGED", "severity_levels": ["warning"] },
   { "event_type": "TEMPLATE_REGISTRY_CHANGED", "severity_levels": ["warning"] },
   { "event_type": "KNOWLEDGE_LIBRARY_CHANGED", "severity_levels": ["warning"] }
 ]
}
```

File: `Axion/libraries/system/registries/notification_destinations.v1.json`

```
{
 "schema_version": "1.0.0",
 "registry_id": "notification_destinations.v1",
 "destinations": [
   { "destination_id": "DEST-CONSOLE", "kind": "console", "enabled": true, "config": {} }
 ]
}
```

File: `Axion/libraries/system/registries/notification_routes.v1.json`

```
{
 "schema_version": "1.0.0",
 "routes_id": "notification_routes.v1",
 "routes": [
   {
     "route_id": "ROUTE-RUNFAIL",
     "match": { "event_type": "RUN_FAILED", "severity": ["error", "critical"] },
     "deliver": { "destination_ids": ["DEST-CONSOLE"] },
     "throttle": { "mode": "dedupe_window", "window_seconds": 300, "dedupe_key":
"event_type_plus_run" }
   },
   {
     "route_id": "ROUTE-GATEFAIL",
     "match": { "event_type": "GATE_FAILED", "severity": ["warning", "error", "critical"] },
     "deliver": { "destination_ids": ["DEST-CONSOLE"] },
     "throttle": { "mode": "dedupe_window", "window_seconds": 300, "dedupe_key":
"event_type_plus_gate" }
   }
 ]
}
```

Piece F — Determinism rules

File: `Axion/libraries/system/SYS-5_determinism_rules.md`

```
---
library: system
id: SYS-5a
schema_version: 1.0.0
status: draft
---
```

SYS-5a — Determinism Rules (Notifications)

- Event payloads must be structured with stable fields (no freeform blobs).
- Routing evaluation is deterministic:
 - evaluate routes in file order
 - stable matching rules
 - stable destination ordering
- Throttle dedupe key must be computable from event fields only.
- Notification emission must be recorded in run logs (event_id, route_id, destination_ids).

Piece G — Validation checklist

File: `Axion/libraries/system/SYS-5_validation_checklist.md`

---
library: system
id: SYS-5b
schema_version: 1.0.0
status: draft

SYS-5b — Validation Checklist

- [ ] event types registry matches schema
- [ ] destinations registry matches schema
- [ ] routes registry matches schema
- [ ] routes reference existing destination_ids
- [ ] throttle mode and dedupe key are valid
- [ ] routing is deterministic (order preserved, stable output)

Policy engine hooks

SYS-6 — Policy engine hooks (risk classes, hard stops, overrides, expiry)

Piece A — Concept and scope

File: `Axion/libraries/system/SYS-6_policy_engine_hooks.md`

---
library: system
id: SYS-6
schema_version: 1.0.0
status: draft

SYS-6 — Policy Engine Hooks

Purpose
`system/` defines how the runtime **invokes** the policy engine in a deterministic way.
The meaning of policies (risk classes, overrides, etc.) lives in `policy/`.
This library defines:
- the hook points (when policy is consulted)
- the request/response schemas
- how decisions are recorded (evidence + expiry)

Hook points (minimum)
Policy must be consulted at:
1) Run start (can this run start?)
2) Pin resolution (are run-level overrides allowed?)
3) Adapter selection (is this execution environment allowed?)
4) Quota enforcement (block vs require approval vs degrade)
5) Gate evaluation overrides (is override permitted, who can approve, expiry required)
6) Kit export (internal vs external constraints)

Determinism requirement
Policy decisions must be:
- computed from explicit inputs (no hidden state)
- recorded in run manifest with decision id + effective policy ids
- stable for the duration of the run

Piece B — Policy hook request schema

File: `Axion/libraries/system/schemas/policy_hook_request.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "policy_hook_request.v1",
  "type": "object",
  "required": [
    "request_id",
    "hook_point",
    "workspace_id",
    "project_id",
    "profile_id",
    "risk_class",
    "pins",
    "context",
    "requested_at"
  ],
  "properties": {
    "request_id": { "type": "string", "pattern": "^POLREQ-[A-Z0-9]{6,}$" },
    "hook_point": {
      "type": "string",
      "enum": [
        "RUN_START",
        "PIN_RESOLUTION",
        "ADAPTER_SELECTION",
        "QUOTA_CHECK",
        "GATE_OVERRIDE",
        "KIT_EXPORT"
      ]
    },
    "workspace_id": { "type": "string", "pattern": "^WS-[A-Z0-9]{6,}$" },
    "project_id": { "type": "string", "pattern": "^PRJ-[A-Z0-9]{6,}$" },
    "profile_id": { "type": "string", "pattern": "^PROFILE-[A-Z0-9_]+$" },
    "risk_class": { "type": "string", "minLength": 2 },

    "pins": {
      "type": "object",
      "description": "Resolved or proposed pins/locks for the run",
      "additionalProperties": true
    },

    "context": {
      "type": "object",
```

```
      "description": "Hook-specific contextual fields (quota exceed, gate id, export target, etc.)",
      "additionalProperties": true
    },

    "requested_at": { "type": "string", "format": "date-time" }
  },
  "additionalProperties": false
}
```

Piece C — Policy hook decision schema (response)

File: `Axion/libraries/system/schemas/policy_hook_decision.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "policy_hook_decision.v1",
  "type": "object",
  "required": [
    "decision_id",
    "request_id",
    "hook_point",
    "outcome",
    "constraints",
    "expiry",
    "decided_at"
  ],
  "properties": {
    "decision_id": { "type": "string", "pattern": "^POLDEC-[A-Z0-9]{6,}$" },
    "request_id": { "type": "string" },
    "hook_point": { "type": "string" },

    "outcome": {
      "type": "string",
      "enum": ["allow", "deny", "require_approval", "degrade"]
    },

    "constraints": {
      "type": "object",
      "description": "Concrete constraints applied (e.g., reduced cap, blocked export, required
approvers)",
      "additionalProperties": true
```

```
    },

    "approval": {
      "type": "object",
      "description": "Present only if outcome=require_approval",
      "required": ["required_roles", "reason"],
      "properties": {
        "required_roles": { "type": "array", "items": { "type": "string" }, "minItems": 1 },
        "reason": { "type": "string", "minLength": 3 }
      },
      "additionalProperties": false
    },

    "expiry": {
      "type": "object",
      "required": ["expires_at"],
      "properties": {
        "expires_at": { "type": "string", "format": "date-time" }
      },
      "additionalProperties": false
    },

    "evidence": {
      "type": "object",
      "description": "Pointers to why/what was evaluated (policy_set_id, rule ids, etc.)",
      "additionalProperties": true
    },

    "decided_at": { "type": "string", "format": "date-time" }
  },
  "additionalProperties": false
}
```

Piece D — Hook integration rules (how runtime uses decisions)

File: `Axion/libraries/system/SYS-6_integration_rules.md`

```
---
library: system
id: SYS-6a
schema_version: 1.0.0
status: draft
```

SYS-6a — Policy Hook Integration Rules

Rule: consult + record
- For each hook point, runtime must:
  1) build a policy_hook_request
  2) evaluate policy (engine)
  3) obtain policy_hook_decision
  4) record decision in run manifest

Rule: stable during run
- Once a decision is recorded for a hook point, it must not change mid-run.
- If expiry is reached mid-run:
  - default behavior is "deny further actions requiring this decision"
  - record a policy_expired event

Rule: outcomes
- allow: proceed
- deny: hard stop
- require_approval: pause run and emit APPROVAL_REQUIRED notification
- degrade: apply constraints and continue (must record what changed)

Gate overrides
- Gate overrides must use GATE_OVERRIDE hook point and must record:
  - gate_id
  - override reason
  - approver identity (when approved)
  - expiry

Piece E — Validation checklist

File: `Axion/libraries/system/SYS-6_validation_checklist.md`

---
library: system
id: SYS-6b
schema_version: 1.0.0
status: draft

SYS-6b — Validation Checklist

- [ ] policy_hook_request matches schema
- [ ] policy_hook_decision matches schema
- [ ] hook_point is one of the defined enum values
- [ ] decisions include expiry and are recorded in run manifest
- [ ] require_approval produces notification + pause semantics
- [ ] degrade records applied constraints deterministically

Minimum viable system

SYS-7 — Minimum viable system set (runtime-required files + “done”)

Piece A — MV system inventory (what must exist for runtime)

File: `Axion/libraries/system/SYS-7_minimum_viable_set.md`

---
library: system
id: SYS-7
schema_version: 1.0.0
status: draft

SYS-7 — Minimum Viable system/ Set

Goal
Define the smallest set of system library assets required to launch and run a pipeline deterministically.

Required runtime files (minimum)
1) Core docs
- SYS-0_purpose.md
- SYS-1_workspace_project_model.md
- SYS-2_pin_lock_policies.md
- SYS-3_adapter_manager.md
- SYS-4_quotas_rate_limits.md
- SYS-5_notification_routing.md
- SYS-6_policy_engine_hooks.md
- SYS-7_minimum_viable_set.md

2) Schemas (must be in schema_registry)
- schemas/workspace.v1.schema.json
- schemas/project.v1.schema.json
- schemas/pin_policy.v1.schema.json
- schemas/pin_set.v1.schema.json
- schemas/capability_registry.v1.schema.json
- schemas/adapter_profile.v1.schema.json
- schemas/command_policy.v1.schema.json
- schemas/quota_set.v1.schema.json
- schemas/quota_profile_modifiers.v1.schema.json (optional)
- schemas/notification_event_types.v1.schema.json

- schemas/notification_destinations.v1.schema.json
- schemas/notification_routes.v1.schema.json
- schemas/policy_hook_request.v1.schema.json
- schemas/policy_hook_decision.v1.schema.json

3) Registries (runtime defaults)
- registries/run_profiles.v1.json
- registries/capabilities.v1.json
- registries/quota_sets.v1.json
- registries/notification_event_types.v1.json
- registries/notification_destinations.v1.json
- registries/notification_routes.v1.json

Required “pin targets” (SYS-2)
At run start, the run manifest must pin:
- adapter_profile_id
- quota_set_id
- policy_set_id
- pin_policy_id
…and record the resolved pin_set.

Required outputs (system responsibilities)
- Deterministic pin_set recorded in run manifest
- Effective quota decision recorded in run manifest
- Adapter capability snapshot recorded in run manifest
- Notification routing decisions recorded in run logs
- Policy hook decisions recorded in run manifest (with expiry)

Piece B — “Definition of done” checklist for system/

File: `Axion/libraries/system/SYS-7_definition_of_done.md`

---
library: system
id: SYS-7a
schema_version: 1.0.0
status: draft

SYS-7a — system/ Definition of Done

system/ is considered “done” when:

Schemas + registries
- [ ] All SYS schemas validate (JSON Schema check)
- [ ] All registries validate against their schemas
- [ ] schema_registry lists all schemas with versions and paths
- [ ] library_index lists all non-schema system files with versions and paths

Runtime behavior (contract-level)
- [ ] Run start resolves pins deterministically into a pin_set
- [ ] Run start resolves effective quotas deterministically and records decision
- [ ] Adapter profile resolves capabilities deterministically and records snapshot
- [ ] Notification routing resolves destinations deterministically and records emission
- [ ] Policy hook requests/decisions are emitted and recorded with expiry

Safety
- [ ] Deny patterns override allow patterns in command policies
- [ ] lock_mode is enforced (open vs pinned_required vs locked)
- [ ] approvals pause run and produce APPROVAL_REQUIRED notifications

Reproducibility
- [ ] A run can be re-executed using only the pinned refs recorded in the manifest

---

Piece C — Minimal folder tree (what must exist on disk)

File: `Axion/libraries/system/SYS-7_minimal_tree.txt`

```
Axion/libraries/system/
  SYS-0_purpose.md
  SYS-0_boundary_checklist.md
  SYS-1_workspace_project_model.md
  SYS-1_determinism_rules.md
  SYS-1_validation_checklist.md
  SYS-2_pin_lock_policies.md
  SYS-2_resolution_rules.md
  SYS-2_validation_checklist.md
  SYS-3_adapter_manager.md
  SYS-3_determinism_rules.md
  SYS-3_validation_checklist.md
  SYS-4_quotas_rate_limits.md
  SYS-4_resolution_rules.md
  SYS-4_validation_checklist.md
  SYS-5_notification_routing.md
```

```
SYS-5_determinism_rules.md
SYS-5_validation_checklist.md
SYS-6_policy_engine_hooks.md
SYS-6_integration_rules.md
SYS-6_validation_checklist.md
SYS-7_minimum_viable_set.md
SYS-7_definition_of_done.md
SYS-7_minimal_tree.txt

schemas/
  workspace.v1.schema.json
  project.v1.schema.json
  pin_policy.v1.schema.json
  pin_set.v1.schema.json
  capability_registry.v1.schema.json
  adapter_profile.v1.schema.json
  command_policy.v1.schema.json
  quota_set.v1.schema.json
  quota_profile_modifiers.v1.schema.json
  notification_event_types.v1.schema.json
  notification_destinations.v1.schema.json
  notification_routes.v1.schema.json
  policy_hook_request.v1.schema.json
  policy_hook_decision.v1.schema.json

registries/
  run_profiles.v1.json
  capabilities.v1.json
  quota_sets.v1.json
  notification_event_types.v1.json
  notification_destinations.v1.json
  notification_routes.v1.json
```

Say **“next”** to start the next library folder. Recommended next is **orchestration/** (ORC-0).